The Witcher Season 3 haulted
The filming has temporarily halted as unconfirmed reports say Geralt tested positive for COVID. **https://dgit.in/sep22-17**

Diablo player smoked $100,000
The player spent $100,000 on the game crafting an OP character and now can't get into matches. **https://dgit.in/sep22-18**

which people don't realise. I'll give you two major ones – power capacity and fiber infrastructure increase for good network signals.

Our telecom infrastructure across the country requires more power to run at optimal levels. The permissions, clearances required for procuring more power are very difficult and also quite expensive. The power that is permitted to be transmitted to telecom towers is one-tenth of international levels, and more towers are difficult to deploy.

The second reason is the need for more fibre towers, and how more towers need to be connected with each other and with the network through fibre so they can carry the capacity of data that they are aggregating on the edge. That requires a huge amount of effort and the government is aware of it. The central government and private industry want to move fast to remove this bottleneck, but local and state governments don't let us put fibre through their territory.

We want more towers to be deployed, and this is where we start. We are expecting about 10 times the speed and capacity of what we have today. But high-speed 5G radio signals do not penetrate walls and therefore require indoor building solutions to be implemented. All of these things are work-in-progress, and there are a lot of innovations which are happening to work around some of these practical 5G deployment problems. A lot of trials are happening and we hope the 5G networks will offer a better overall experience than 4G or before.

Q How will 5G roll-out across India impact us positively?

A While e-commerce takes over 78 per cent of the traffic on any network across all Indian telecom operators, the bandwidth requirement of e-commerce traffic is actually very low compared to video streaming or online gaming. So e-commerce in terms of 5G will be an application that requires tele-density, but will not affect the speed and latency that will be required in different areas. Use cases like 3D holograms or teleporting your e-presence or even online gaming will get some boost. But something like tele-medicine services will seriously ramp up across the length and breadth of the country thanks to 5G.

Another thing that will be grappled quickly due to 5G deployment is a merger of telecom, electronics and computing to be treated as one silo – it can no longer be looked at separately from a legal, regulatory and policy framework, which currently is not happening.

Cyber security has still been outside the domain of the telecom sector at the core infrastructure level, but that also needs to be worked into as a fundamental pillar of network security. It cannot be an afterthought anymore. The work has to be started from the drawing board, on both the telecom technology as well as cyber security.

Q How will indigenous Indian 5G hardware ramp up to safeguard our 5G networks against Chinese ODMs?

A The Indian government has already taken a slew of measures which will ensure that this (deployment of indigenous Indian telecom hardware) happens as fast as possible. For now, the components which have been used in our telecom network can be divided into passive and active. Passive isn't a problem, we import industry-standard hardware from outside but yes, from the Make in India perspective, those components should also be made in India.

The active network needs software plus hardware stack expertise. The software component, India doesn't have a problem with. We have always been making software for the entire world, and we can do it. But we can't yet do both hardware and software, because the hardware piece of the puzzle is difficult to crack in a hurry. Designing and manufacturing of semiconductor chips requires massive investment, and it's not going to happen overnight. Currently, a lot of emphasis is being given on importing knowledge, skills, and expertise to get these semiconductor companies to manufacture in India. It's being incentivized by the government too. However, for now sensitive telecom components will need to be imported from abroad. The Indian government has also made a conscious decision to continue with the past practice of using international standard based telecom equipment so that we are not depending on any one country for our needs. Of course, we are not satisfied with that, we want an ecosystem to be created in India, where we can manufacture, we can design, and consume in India and export outside to the world.

The advantage of China was cheap labour and economies of scale. We have cheap labour. What we don't have is economies of scale. We don't have a manufacturing base, so incentives have now been offered to these companies to come to India, and start making in India. China has been taken out of the ecosystem in many parts of the telecommunications market, and our strategy is to not only make in India for Indian telecom needs but also to eventually become a net exporter. It will take some time. These steps are being monitored at the highest levels of the Indian government, because of what's at stake. So there's reason to believe the implementation of any indigenous hardware in the telco stack will also be faster in the future.

Q How 5G will help Indian telcos evolve in the near future?

A I can make a small prediction here: The telecom business of providing networks is not going to be a paying business anymore. Right. You will eventually find that newer business verticals will be developed by these Indian telecom operators themselves – not just for large businesses, but even for us normal citizen consumers. Customers will be increasingly offered consolidated 360-degree package solutions, because these offerings will come from telcos themselves who understand business on one hand and 5G consumption on their network. The hope is that enough consumers will find these offerings to be transformative enough within the next year or so.